Run report: run-20261012T0000Z-binary-scan

What ran. `./target/release/examples/binary_gls_scan --m M --dmax X --keep 6 --samples 5000 --lmax 61 --json …` with these parameters:

M	X	start
127	4,000,000	00:00:27Z
83	8,000,000	00:00:37Z
131	8,000,000	00:00:46Z

Final source, before the checkpoint commit after 8b58692c5. rustc 1.97.0, release, cloud container, not isolated, seed 20261012. Wall times: 5.2 s, 9.6 s and 10.2 s (`../../binary_scan_m*.log`).

Method. For each fundamental D ≡ 1 (mod 8) the scan requires:
- the class of $\mathfrak{p}_2$ has order exactly m;
- Cornacchia gives $4\cdot 2^m = t^2 + |D|v^2$ with t odd;
- $r = ((2^m - 1)^2 + t^2)/2$ is a probable prime.

It then searches odd prime $\ell \le 61$ for $[\mathfrak{l}][\mathfrak{p}_2]^k = 1$. It keeps F-oriented $\alpha = (A + B\sqrt{D})/2$ of norm $\ell \cdot 2^{\min(k,\, m-k)}$ and ranks them by model cost 7.5ℓ + 2·(F steps) M. GLS-2 and GLV-GLS-4 decompositions are measured on 5,000 seeded scalars.

Runs requested / completed. 3 / 3, exit 0. An earlier m = 127 run of the same scan printed “kept 2 with prime r” while listing 3 prime-r rows; the wording conflated two counts. The summary line was corrected to report both counts, and the run was repeated. Only the repeat is recorded.

m	scanned	ord = m	prime r	with loop	cheapest (D: ℓ, F steps, M)
83	810,576	440	4	0	—
127	405,292	319	3	2	−1532327: 3, 53, 128.5
131	810,576	459	4	3	−283583: 13, 16, 129.5

GLV-GLS-4 max coefficient bits: 64 (m = 127) and 66 (m = 131), against GLS-2 at 126 and 130.

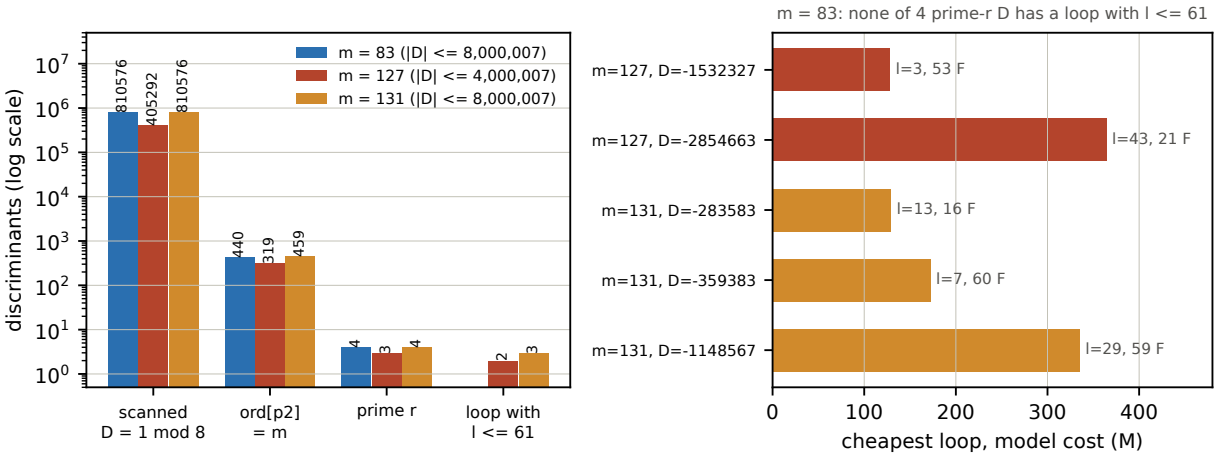


Figure 1: Recorded funnel counts per m (log scale), and the cheapest loop’s model cost per kept discriminant (run-20261012T0000Z-binary-scan).

Observations only. Interpretation: `../../README.md` §4.